Quantification: all 7 images
==========
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Columns 1 to 6 of 35

| image_label | lacuna count | interior lacuna count | median lacuna area (px^2) | roots per cell | ring length 30 px per cell (px) | ring length 60 px per cell (px) |
|---|---|---|---|---|---|---|
| 535_z13c1-2 | 23 | 20 | 1363.5 | 7.8 | 237.25 | 472.15 |
| 535_z27c1-2 | 24 | 20 | 1965 | 7.3 | 228.95 | 503.8 |
| 536-63x_Hyp_RD_z03c1-2 | 22 | 18 | 2332 | 10.7222 | 300.167 | 582.611 |
| 536-63x_Hyp_RD_z08c1-2 | 22 | 21 | 2738 | 10.619 | 316.952 | 632.143 |
| 536-63x_Hyp_RD_z15c1-2 | 11 | 10 | 2611.5 | 9.3 | 315.9 | 812.7 |
| I534_z11c1-2 | 36 | 29 | 1735 | 10.3103 | 281.759 | 519.897 |
| I534_z25c1-2 | 32 | 21 | 2037 | 7.2381 | 237.19 | 489.762 |

Columns 7 to 12 of 35

| image_label | field length density (px^-1) | perimeter per cell (px) | ring area 30 px per cell (px^2) | ring area 60 px per cell (px^2) | in-frame fraction 30 px per cell | in-frame fraction 60 px per cell |
|---|---|---|---|---|---|---|
| 535_z13c1-2 | 0.0210406 | 257.918 | 8353.1 | 17278.9 | 1 | 0.9977 |
| 535_z27c1-2 | 0.0248805 | 284.056 | 8979.75 | 19216.7 | 0.9934 | 0.9872 |
| 536-63x_Hyp_RD_z03c1-2 | 0.0220253 | 307.404 | 9176.78 | 20707.3 | 0.9793 | 0.9657 |
| 536-63x_Hyp_RD_z08c1-2 | 0.0269717 | 326.693 | 9467.24 | 20555.6 | 0.9877 | 0.9754 |
| 536-63x_Hyp_RD_z15c1-2 | 0.0259634 | 320.57 | 10797.3 | 25631.5 | 0.9949 | 0.9787 |
| I534_z11c1-2 | 0.0284684 | 253.534 | 8504.72 | 16527.5 | 0.9888 | 0.9705 |
| I534_z25c1-2 | 0.0257757 | 241.931 | 8563.19 | 18199.7 | 1 | 0.9929 |

Columns 13 to 18 of 35

| image_label | ring density 30 px per cell (px^-1) | ring density 60 px per cell (px^-1) | roots per 100 px perimeter per cell | ring attached length 30 px per cell (px) | ring attached length 60 px per cell (px) | ring length weighted 30 px per cell (px) |
|---|---|---|---|---|---|---|
| 535_z13c1-2 | 0.0267732 | 0.0261594 | 3.1259 | 205.6 | 360.6 | 267.268 |
| 535_z27c1-2 | 0.0261157 | 0.0266667 | 2.7197 | 190.9 | 326.55 | 259.083 |
| 536-63x_Hyp_RD_z03c1-2 | 0.0336878 | 0.0296449 | 3.6728 | 267.389 | 432.056 | 339.396 |
| 536-63x_Hyp_RD_z08c1-2 | 0.0321568 | 0.0295088 | 3.301 | 286.333 | 498.381 | 358.136 |
| 536-63x_Hyp_RD_z15c1-2 | 0.0291845 | 0.0314224 | 2.9425 | 277.2 | 650.8 | 356.961 |
| I534_z11c1-2 | 0.0337116 | 0.0319368 | 4.1779 | 255.345 | 412.621 | 319.673 |
| I534_z25c1-2 | 0.0283096 | 0.0277136 | 2.9622 | 202.095 | 334.381 | 269.773 |

Columns 19 to 24 of 35

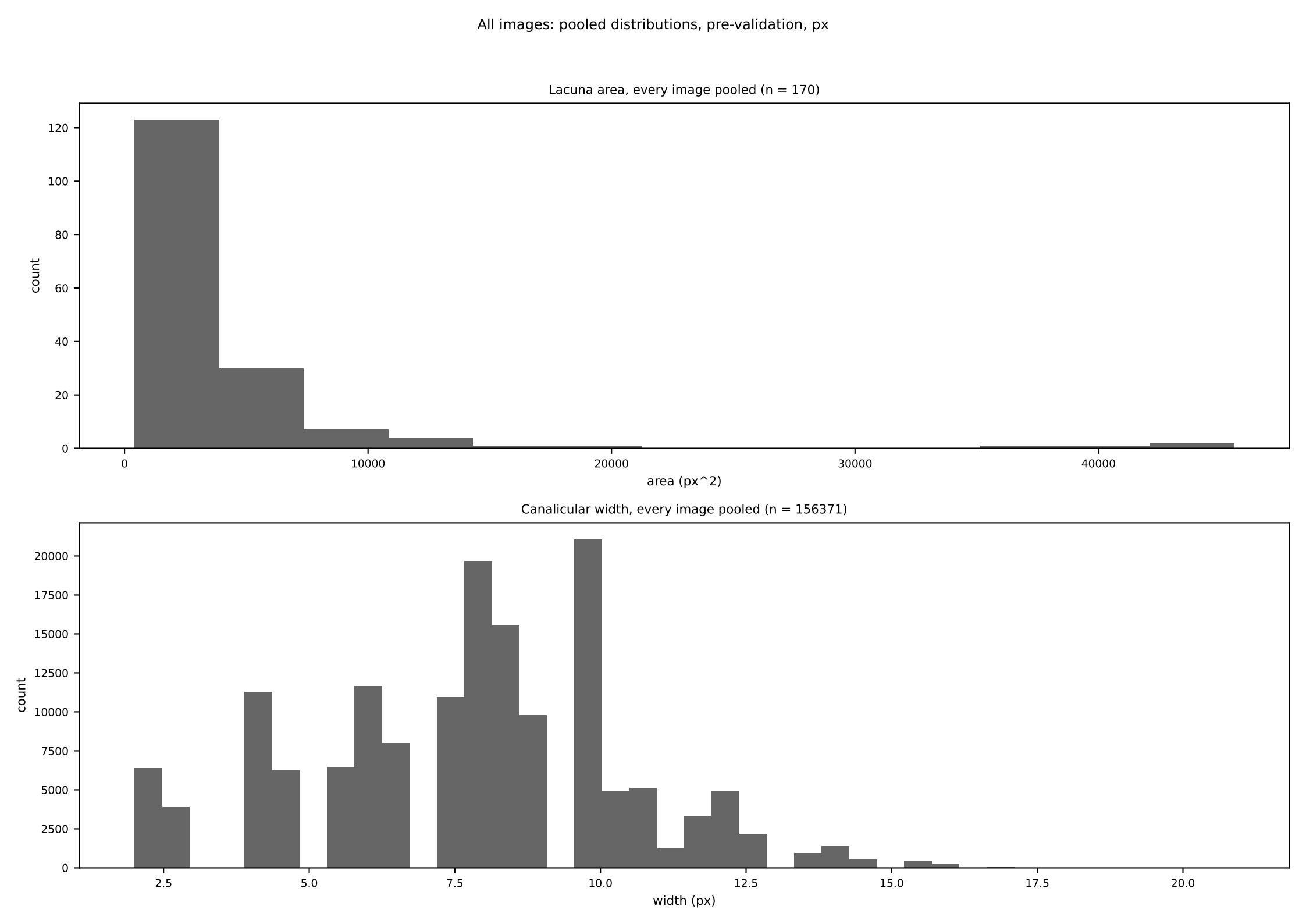
| image_label | ring length weighted 60 px per cell (px) | Sholl crossings 10 px per cell | Sholl crossings 20 px per cell | Sholl crossings 30 px per cell | field length density weighted (px^-1) | field length density without flagged regions (px^-1) |
|---|---|---|---|---|---|---|
| 535_z13c1-2 | 533.236 | 6 | 6 | 6.8 | 0.023686 | 0.0205717 |
| 535_z27c1-2 | 569.974 | 6.05 | 6.7 | 7.15 | 0.0280576 | 0.0233366 |
| 536-63x_Hyp_RD_z03c1-2 | 658.465 | 7.5556 | 7.5556 | 7.5556 | 0.0248877 | 0.0224018 |
| 536-63x_Hyp_RD_z08c1-2 | 719.414 | 8 | 8.4286 | 8.2381 | 0.0306341 | 0.0270237 |
| 536-63x_Hyp_RD_z15c1-2 | 921.41 | 7.5 | 10.7 | 11.4 | 0.0294714 | 0.0261397 |
| I534_z11c1-2 | 590.198 | 6.2759 | 7.1379 | 7.6897 | 0.0322797 | 0.0298895 |
| I534_z25c1-2 | 557.308 | 5.7619 | 6.5238 | 7.1429 | 0.0293041 | 0.0268387 |

Columns 25 to 30 of 35

| image_label | field length density in ROI (px^-1) | loops per cell | share of a cell's edges on a loop | loops per field | share of field edges on a loop | canalicular width mean (px) |
|---|---|---|---|---|---|---|
| 535_z13c1-2 | | 1.25 | 0.1166 | 29 | 0.0852 | 7.3607 |
| 535_z27c1-2 | | 1.1 | 0.0632 | 31 | 0.0924 | 7.5882 |
| 536-63x_Hyp_RD_z03c1-2 | | 1.0556 | 0.0778 | 31 | 0.099 | 7.5169 |
| 536-63x_Hyp_RD_z08c1-2 | | 2.2381 | 0.1541 | 58 | 0.1613 | 7.8468 |
| 536-63x_Hyp_RD_z15c1-2 | | 5.2 | 0.213 | 59 | 0.1803 | 7.7509 |
| I534_z11c1-2 | | 1.3793 | 0.099 | 56 | 0.1409 | 8.2307 |
| I534_z25c1-2 | | 1.7143 | 0.1232 | 45 | 0.1275 | 8.015 |

Columns 31 to 35 of 35

| image_label | canalicular width median (px) | canalicular width p10 (px) | canalicular width p90 (px) | canalicular width pixels used | canalicular width in ring 30 px per cell (px) |
|---|---|---|---|---|---|
| 535_z13c1-2 | 8 | 4 | 10.7703 | 19466 | 7.2325 |
| 535_z27c1-2 | 8 | 4 | 10.7703 | 18898 | 7.3625 |
| 536-63x_Hyp_RD_z03c1-2 | 8 | 4 | 10.7703 | 21354 | 7.3319 |
| 536-63x_Hyp_RD_z08c1-2 | 8 | 4 | 10.7703 | 25291 | 7.5814 |
| 536-63x_Hyp_RD_z15c1-2 | 8 | 4 | 10.7703 | 25264 | 7.8457 |
| I534_z11c1-2 | 8.2462 | 4 | 11.6619 | 23613 | 7.832 |
| I534_z25c1-2 | 8.2462 | 4 | 11.3137 | 22485 | 7.5731 |


All images: pooled distributions, pre-validation, px
Lacuna area, every image pooled (n = 170)
count
area (px^2)
0
20
40
60
80
100
120
0
10000
20000
30000
40000
Canalicular width, every image pooled (n = 156371)
count
width (px)
0
2500
5000
7500
10000
12500
15000
17500
20000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0